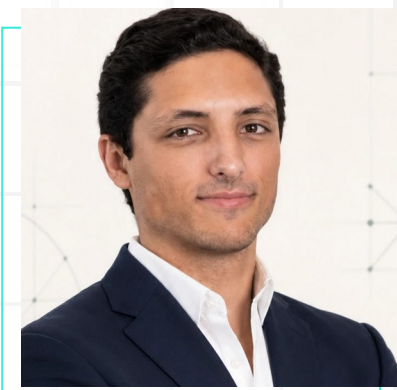


From Design to Digital Twin: Physics and AI for Fatigue-Aware Floating Offshore Wind Towers

From Knowledge to Competitiveness

Portugal-US Partnerships 2026 Summit

MIT Portugal Program



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The challenge

Floating turbines above 20 MW must prove 25 years of tower fatigue life, with thousands of simulations per design. Once installed, tower loads are rarely measured along the tower. For Portugal's deep Atlantic waters, solving both is key to a competitive offshore wind sector.

Our approach

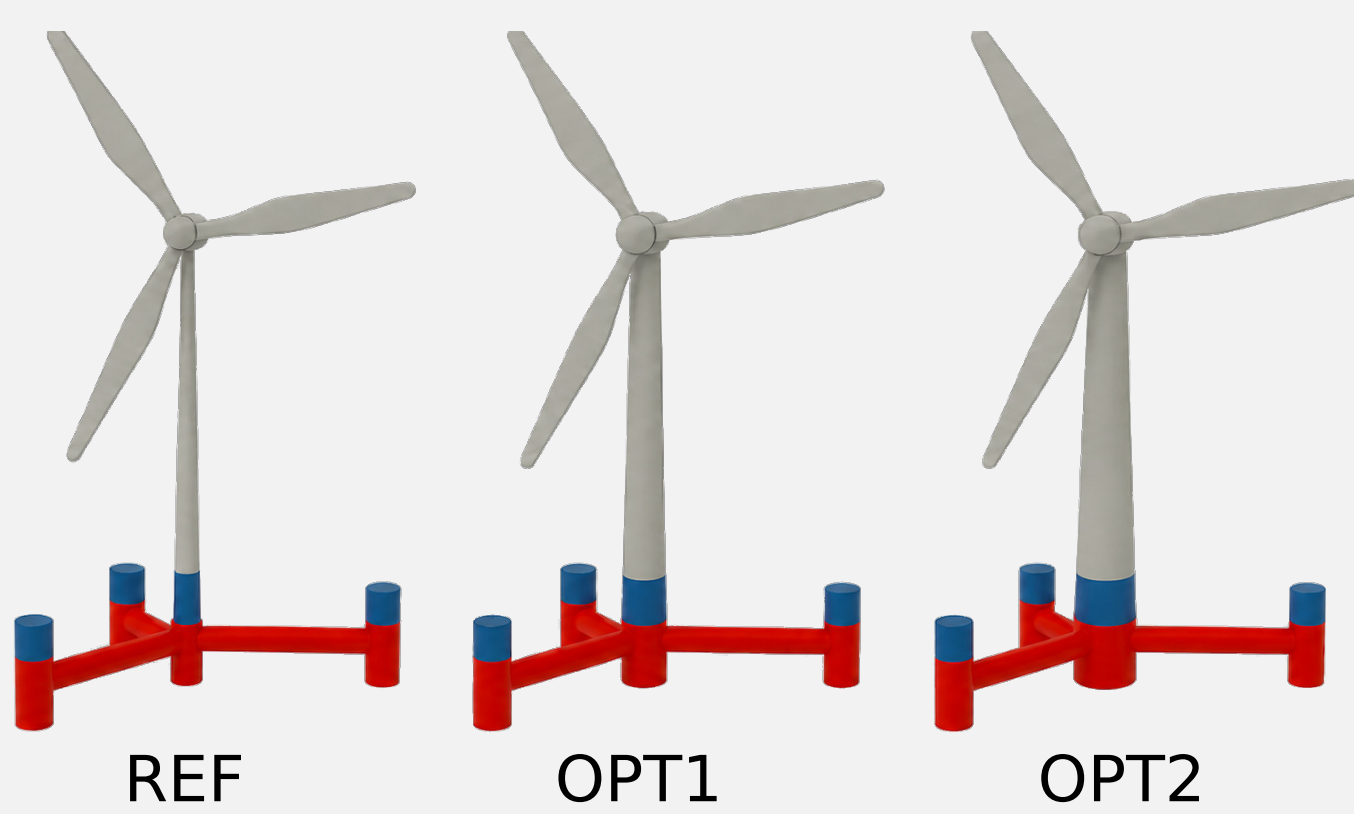
One family of physics + AI tools on the IEA 22 MW floating reference turbine: 1 design → 2 benchmark → 3 simulate less → 4 monitor.

The same structural physics governs design, simulation, learning and monitoring.

Physics-Guided Design

AI Benchmarks and Operation

1 FLOAT: A 25-Year Tower [2]



Fatigue life of the tower

IEA reference

~9 months

FLOAT tower (OPT1/OPT2)

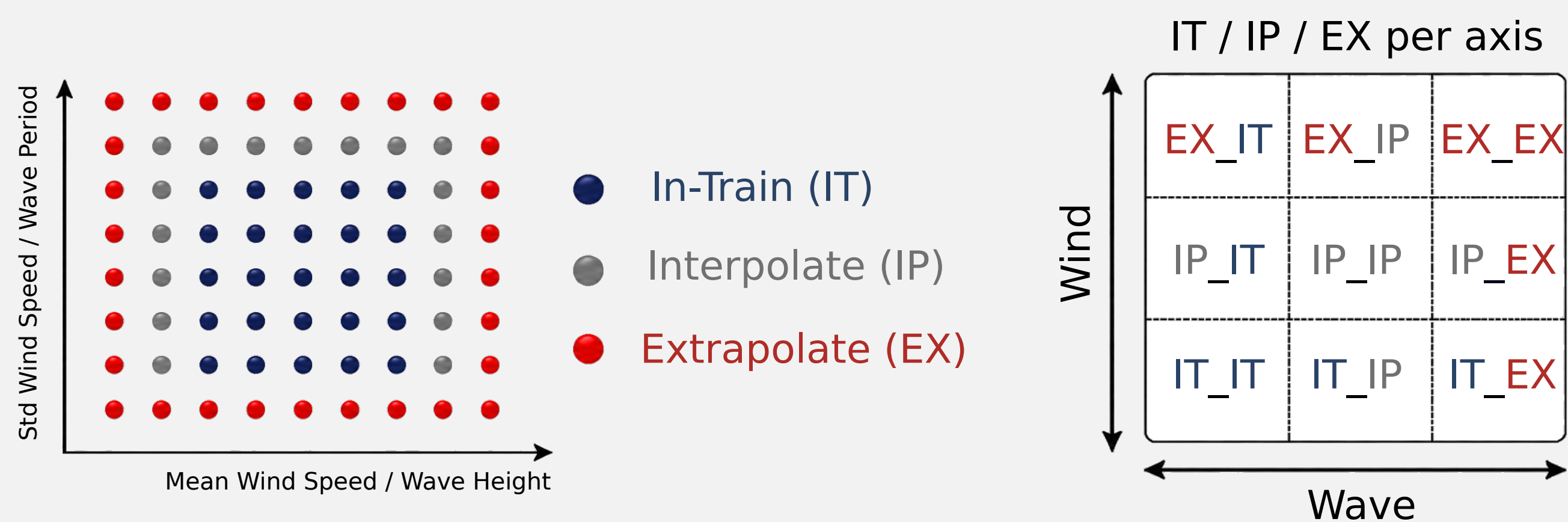
25 years

25-year design life

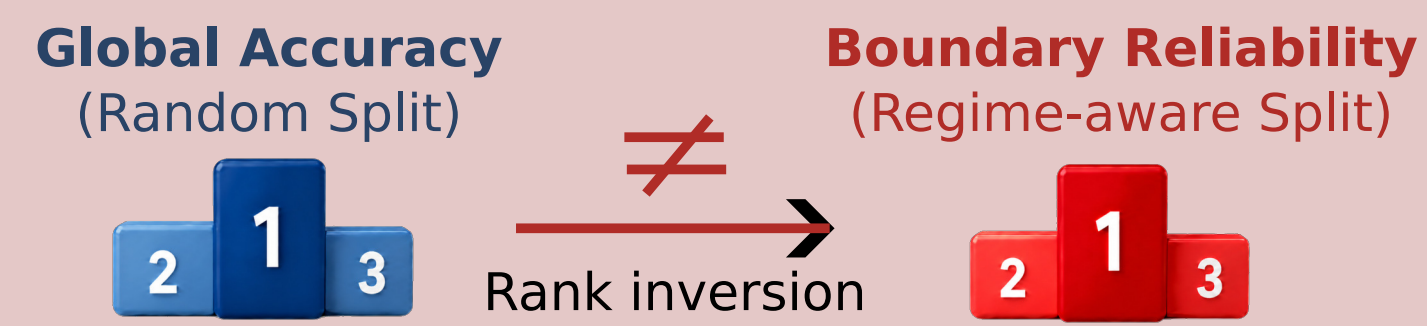
6,468 simulations per tower (DLC 1.2); rainflow, S-N and Miner fatigue; verified by full re-simulation

Merged into the official IEA 22 MW reference model

2 FLOATBench: Dataset & Benchmark [3]



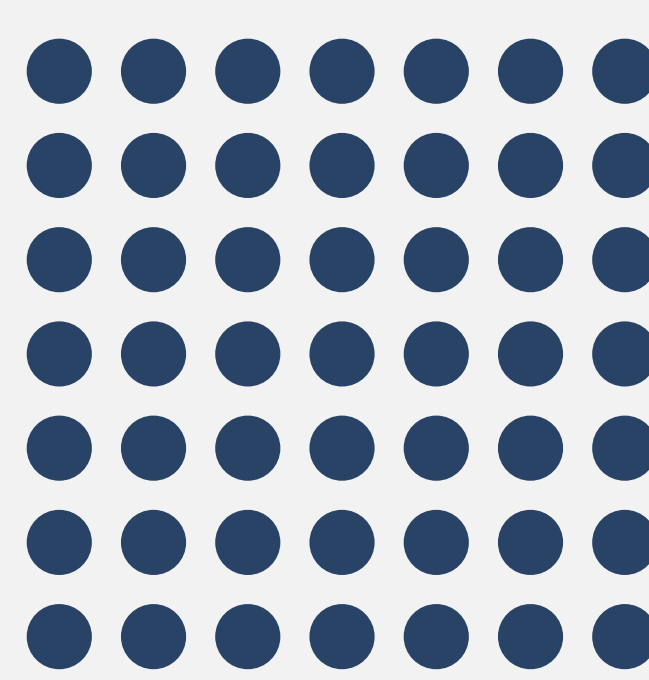
582,120 samples labelled with damage



Rankings invert random vs regime-aware splits

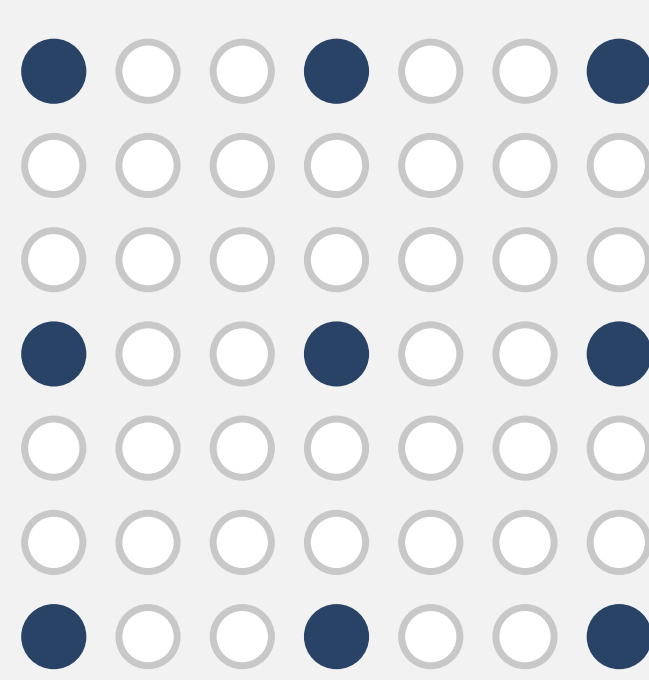
3 NeuroFLOAT: 5.4× Fewer Runs [4]

Full campaign



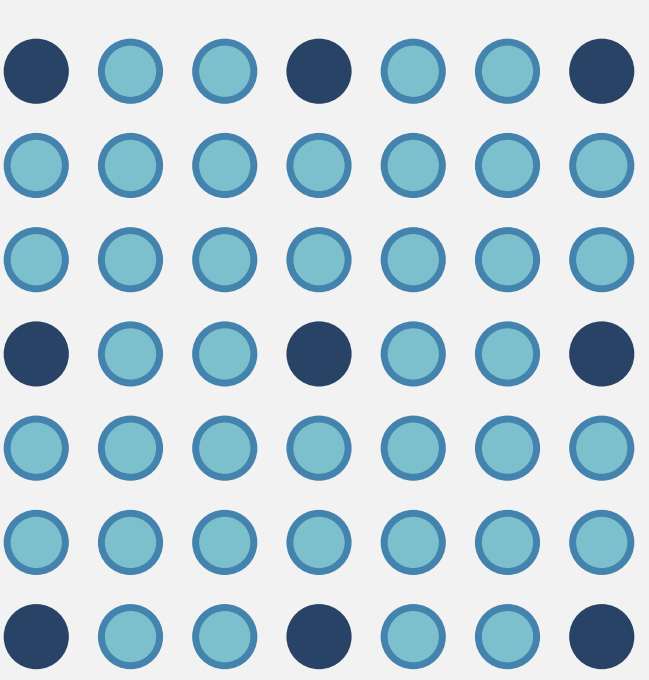
7×7 wave plane
6,468 runs

Physics screening



3×3 subset
1,188 runs

AI reconstruction

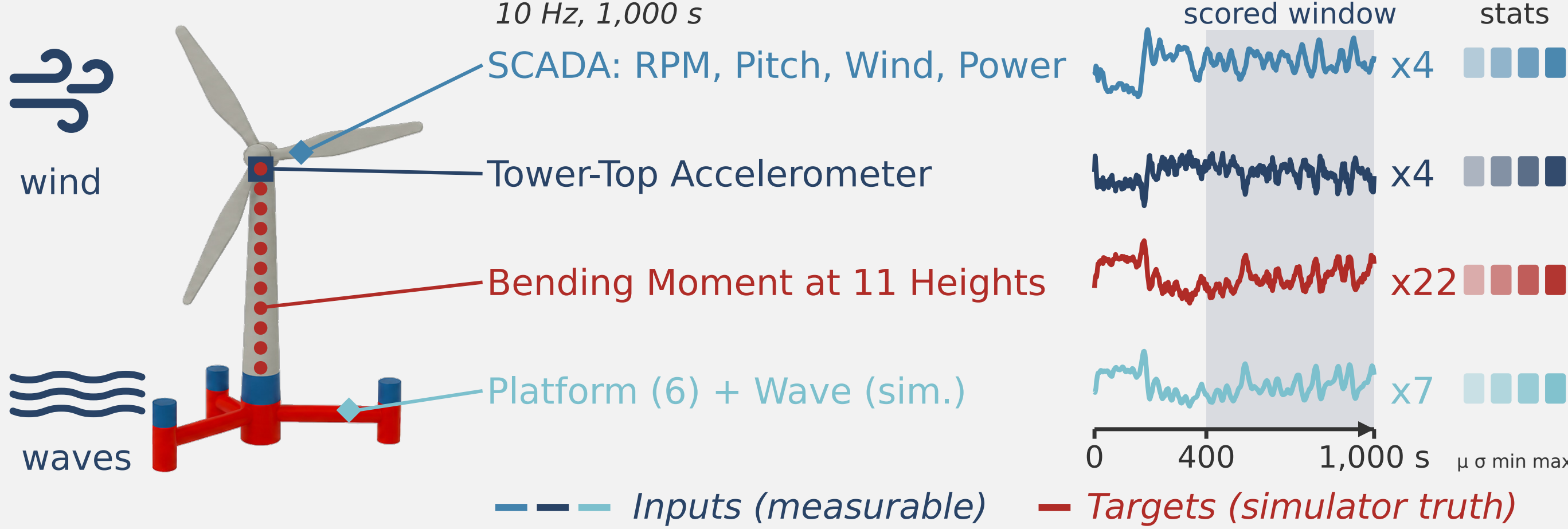


surrogate fills omitted states

● simulated ● predicted by AI

5.4× fewer runs, damage within ~0.5% of full campaign

4 FLOATSense: Virtual Sensing [5]



Tower-top accelerometer + SCADA → fore-aft moment at 11 heights; physics baseline + 19 learned models

Height matters: the base winner can fail at the top

For industry

Before installation: a faster design loop with far fewer simulations per concept. After installation: tower loads from nacelle sensors (simulation-validated; field next).

Portugal-US partnership

FEUP and Univ. of Aveiro with MIT and Brown, TU Delft and the IEA Wind community. Continued in the MIT Portugal Seed Project 2024.14532.MIT (Aveiro and MIT).

Take-home

Physics provides reliability, AI provides scale.

Next: generative design and physics-informed digital twins.

References: [1] Alves Ribeiro et al., Applied Energy 397 (2025) 126294. [2] FLOAT, Ocean Engineering 362 (2026) 126285. [3] FLOATBench, [4] NeuroFLOAT, [5] FLOATSense, under review. MIT Portugal Seed Project, doi.org/10.54499/2024.14532.MIT.

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